BRENTON YASUOKA

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EDUCATION

Colorado State University May 2026
B.S. Mechanical Engineering & B.S. Biomedical Engineering | Dean's List

PROFESSIONAL EXPERIENCE

Automation and Controls Engineer | Multi-Dimensional Integration Jul 2026 - Present
- Program, commission, and troubleshoot ControlLogix PLCs, PowerFlex VFDs, HMIs, remote I/O, IO-Link devices, and EtherNet/IP networks using Studio 5000.
- Lead on-site functional testing and startup, resolving controls, wiring, and communication issues through production handoff.
- Coordinate engineers, technicians, vendors, and customers as Controls Commissioning Site Lead, communicating technical risks, issues, and schedule impacts.

Automation Engineer Intern | Kodak Moments-Alaris Jun 2025 - Sep 2025
- Developed a production-ready thermal ribbon respooling system integrating mechanical, electrical, and controls hardware from prototype through deployment.
- Programmed Allen-Bradley PLCs and HMIs for pneumatic, vacuum, and motion subsystems, achieving 5 mm positioning precision over 10 m using encoder feedback.
- Collaborated with 6 engineers and 3 suppliers to build a $2,500 quality-control system replicating key functionality of a $500K production machine.
- Designed mechanical components for machinability and coordinated with suppliers.

Research Assistant | Electro Anaerobic Digestion Project Jun 2024 - Jan 2025
- Developed a GUI and macro system to automate daily resets and data logging, eliminating manual intervention in an experimental workflow.
- Designed a real-time pH control system to maintain target microbial conditions across 4 H-cells.

Undergraduate Teaching Assistant | Colorado State University Aug 2022 - May 2026
- Led MATLAB and Arduino instruction and provided technical support for 400+ students and 6 faculty; trained Learning Assistants and mentored students.

President | Biomedical Engineering Society Jun 2024 - May 2025
- Led six officers and secured sponsorships, tours, and mock interviews for 200+ students.

SELECTED PROJECTS

Project Lead | CASE Adaptive Vehicle Controls May 2022 - Aug 2025
- Led a 10+ member engineering team to design and fabricate adaptive vehicle controls, coordinating CAD, machining, additive manufacturing, accessibility, and integration.
- Engineered a custom vehicle control panel around accessibility, mounting, affordability, and compatibility with existing vehicle components.

Python Engineering Analysis
- Developed Python scripts using pandas and NumPy to parse LabVIEW measurements, calculate stress and strain, and export CSV and Excel analysis outputs.
- Grouped measurements by rounded strain values and generated trial and summary workbooks to support comparison of repeated tests.

TECHNICAL SKILLS

INTEGRATION & TEST: Commissioning, functional testing, hardware/software troubleshooting
PROGRAMMING & DESIGN: Python, pandas, NumPy, MATLAB, SolidWorks, AutoCAD
CONTROLS: Studio 5000, ControlLogix, PLC/HMI programming, PowerFlex VFDs, EtherNet/IP, IO-Link